# NRSC Bulletin No. 2012-001
# Wireless Outages
# December 2012

## *Background*

The Alliance for Telecommunications Industry Solutions (ATIS) Network Reliability Steering Committee (NRSC) responded to the Federal Communications Commission (FCC) Public Safety and Homeland Security Bureau's (PSHSB) interest in a perceived increase in wireless outages.

The Wireless Outages Subteam was initiated in January 2012 to address data presented to the NRSC by the FCC at the December 12, 2011, NRSC public meeting showing that the frequency of Network Outage Reporting System (NORS) outage reports for the wireless category was outside statistical control limits, and that the frequency trend line was increasing.

## *Methodology of the Wireless Subteam*

Members of the Subteam consisted of representatives of the NRSC member companies and data analysis was performed by a representative of Applied Communication Sciences. The study included a series of analyses of wireless outage data provided by NRSC wireless carriers, taking into account sympathy and non-sympathy reports. This data was comprised of all final NORS reports filed by the participating wireless carriers with 900,000 Wireless User Minutes cited as the Reason Reportable during the 31-month period from January 2010 through July 2012.

## *Findings of the Wireless Subteam*

The Subteam observed a significant downward trend of outage reports beginning in November 2011. Participants analyzed a range of data for the study period to determine if that recent downward trend had stabilized. Further, the team analyzed outage durations to investigate whether the duration of these types of outages were increasing. Ultimately it was determined that the duration does not appear to be increasing.

The Subteam found that 65% of the outage reports were sympathy reports (outages that "occurred in another company's network") and that the leading cause of those outages was cable damage. However, the Subteam also completed a thorough analysis of keywords, excluding sympathy reports, covering both direct and root causes. This resulted in a unified cause trend analysis that identified the number one cause of outages across the industry is in the *Procedural-Service Provider-Other* cause group. Further analysis of the key words included in the description of these procedural outage events showed that 88% of these events contained the words "planned" or "maintenance," indicating that these events likely occurred during a maintenance window.

## *Recommendation of the Wireless Subteam*

As new technology is deployed and consumer demand continues to increase, the need for more planned maintenance will also increase. The NRSC recommends that the FCC consider revising the Part 4 outage reporting requirements associated with planned activity during designated maintenance windows. One option would be to reduce the number of subscribers reported as affected by the outage. This could be accomplished by reducing the concentration ratio which would give a truer picture of the actual impact of the outage.[1] Communication providers

[1] Section 4.9 (b) of the Commission's Rules state that "in determining the number of users potentially affected by a failure of a switch, a concentration ratio of 8 shall be applied."

select the maintenance window based on the measured traffic handled by the specific location in order to minimize subscriber impact. As communication providers continue to expand and upgrade their networks and services, it is incumbent upon them to take all reasonable steps to minimize the impact and duration of such outages. The data indicates that 24% of the planned maintenance activity had some type of procedural or human error associated with it. The Wireless Subteam also identified that the second largest root and direct causes of outages were *Hardware Failure Other* and *Circuit Pack/Card Failure-Other.* During the study period, these cause codes increased at a significant rate; however, the overall outage frequency did not increase at a significant rate.[2] Additional analysis of these categories concluded there is no specific equipment type that is causing the problem. The NRSC believes the following Best Practices will help ensure that outage durations associated with the issues identified in this bulletin are kept to a minimum.

| Number | Description |
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| 8-5-0536 | As appropriate, Network Operators and Service Providers should deploy security and reliability related software updates (e.g., patches, maintenance releases, dot releases) when available between major software releases. Prior to deployment, appropriate testing should be conducted to ensure that such software updates are ready for deployment in live networks. Equipment Suppliers should include such software updates in the next generic release and relevant previous generic releases. |
| 8-6-5073 | Network Operators, Service Providers and Equipment Suppliers should perform risk assessment on significant network changes (e.g., technology upgrades). |
| 8-7-0413 | Maintenance Notification: Network Operators and Service Providers should communicate information on service affecting maintenance activities and events to their customers, as appropriate. |
| 8-7-0418 | Back-out MOPs: Network Operators and Service Providers should, where appropriate, have a documented back-out plan as part of a Method of Procedure (MOP) for scheduled and unscheduled maintenance activities. |
| 8-7-0501 | Network Operators and Service Providers should report problems discovered from their operation of network equipment to the Equipment Supplier whose equipment was found to be the cause of problem. |
| 8-7-0618 | Network Operators and Service Providers should establish mutually agreed upon reliability thresholds with Equipment Suppliers for new hardware (e.g., routers, switches, call servers, signaling servers) brought into service on the network. |
| 8-7-0510 | Network Operators, Service Providers and Equipment Suppliers should, by design and practice, manage critical Network Elements (e.g., Domain Name Servers, Signaling Servers) that are essential for network connectivity and subscriber service as critical systems (e.g., secure, redundant, alternative routing). |
| 8-7-0559 | Service Providers and Network Operators should consider validating upgrades, new procedures and commands in a lab or other test environment that simulates the target network and load prior to the first application in the field. |
| 8-7-0455 | Equipment Suppliers should consider a program to remove cards or modules from circulation that have a history of failure even if tests indicate "No Trouble Found." |
| 8-7-0504 | Network Operators and Service Providers, in order to facilitate asset management and increase the likelihood of having usable spares in emergency |

[2] For additional information, refer to the NRSC Outage Reporting Advisory Subcommittee (ORAS) examination of hardware failure cause codes based on outage reports coded as “Hardware Failure-Other.”

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| | restorations, should consider maintaining "hot spares" (circuit packs electronically plugged in and interfacing with any element management system, as opposed to being stored in a cabinet) for mission critical elements. |
| 8-7-5083 | Network Operators, Service Providers and Equipment Suppliers should maintain the availability of spares for critical network systems. |
| 8-7-5084 | Hardware & Software Quality Assurance: Network Operators, Service Providers and Equipment Suppliers should consider ensuring that outsourcing of hardware and software includes a quality assessment, functional testing and security testing by an independent entity. |
| 8-7-0565 | Equipment Suppliers should establish and use metrics to identify key areas and measure progress in improving quality, reliability, and security during product development and field life cycle. |

The most up-to-date (ATIS) Best Practices can be found at:

- https:// www.atis.org/bestpractices/Default.aspx
- https://www.fcc.gov/nors/outage/bestpractice/BestPractice.cfm

## *Conclusion*

The Wireless Subteam determined that Procedural Service Provider and Hardware Failure were the top two contributors to the increase of wireless outages between May 2010 and July 2012. As such, the NRSC recommends that communications providers review the use of the Best Practices listed above and consider expanding implementation as appropriate. The NRSC believes this will have a positive impact on the wireless outage trend; however, it is likely that the number of wireless outages related to planned activity, as reported under the current rules, will remain high, as wireless providers continue to upgrade equipment in the field to meet customer demand for new services and features. Therefore, the NRSC recommends that the FCC consider revising the outage reporting requirements associated with planned activity during designated maintenance windows. The NRSC believes doing this will provide a truer picture of actual customer impact.